IE 301 · OPERATIONAL ANALYSIS OF MULTIMODAL INTERCHANGE HUBS

Kanjurmarg Station: Stochastic Simulation of Foot Overbridge Utilization Under Increased Passenger Demand

Technical report — five-gate model, Ghatkopar-referenced demand scenarios, utilization results and limitations

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IE 301 project team · October 2026 · Version 3 (revised route splits)

FOB 2 baseline utilization	FOB 3 baseline utilization	Busiest simulated minute
20.7%	**9.3%**	**45.1%**
→ 31.9% with +100 pax/min	→ 13.8% with +100 pax/min	FOB 2, +100 pax/min (of reference capacity)

Utilization = one-minute FOB crossings ÷ reference capacity (W × 60 pax/m/min); mean of 30 stochastic replications. Gate route splits are modelling estimates (with random noise), the East-side split and the 500 pax/min Ghatkopar reference are assumptions, and the +50/+100 pax/min scenarios are input rates, not ridership forecasts. Simplified conceptual layout — not to scale.

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1. Executive Summary

A stochastic AnyLogic model estimates FOB 2 and FOB 3 utilization at Kanjurmarg for three scenarios: baseline, **+50** and **+100 pax/min** of extra entering demand (10% and 20% of an assumed 500 pax/min Ghatkopar flow; 80% via Huma, 20% via Gandhi Nagar). Five gates, FOB 1 closed, Poisson arrivals, gate route splits with ±3 pp noise, 30 replications × 60 min.

Table 1. Mean one-minute utilization (% of reference capacity, ± 95% CI).

Scenario	FOB 2	FOB 3
Baseline	20.7 ± 0.5	9.3 ± 0.4
+50 pax/min	26.3 ± 0.5	11.6 ± 0.4
+100 pax/min	31.9 ± 0.6	13.8 ± 0.5

- FOB 2 is about twice as loaded as FOB 3 in every scenario. +100 pax/min adds about 11 points on FOB 2 and 4 on FOB 3.
- Busiest single minute in any run: 45.1% (FOB 2, +100 pax/min). Both bridges stay below reference capacity.
- Expected baseline flows from the assumed splits (68.9 and 32.6 pax/min) are close to the observed 70.0 and 33.3 pax/min. This is a consistency check, not validation.

2. Problem Definition and Objectives

Kanjurmarg is expected to become an interchange with Metro Lines 4 and 6. With FOB 1 closed, bridge movement relies on FOB 2 and FOB 3. Questions:

1. How is gate demand split between direct PF1/PF1A, FOB 2, FOB 3 and the skywalk?
2. What are baseline FOB 2 and FOB 3 utilizations?
3. What changes with +50 and +100 pax/min?
4. How variable are the results?
5. How do estimated FOB flows compare with observed counts?
6. Does extra demand change the FOB 2 / FOB 3 balance?

Scope note. +50 and +100 pax/min are scenario inputs, not ridership forecasts.

3. Scope and Simplified Station Layout

West side (north → south): Naval Colony gate, FOB 2, Huma gate, FOB 3, Gandhi Nagar gate. FOB 1 is closed. An east-side **skywalk connects FOB 2 and FOB 3** (connector, not an FOB) with Northeast and Southeast exits.

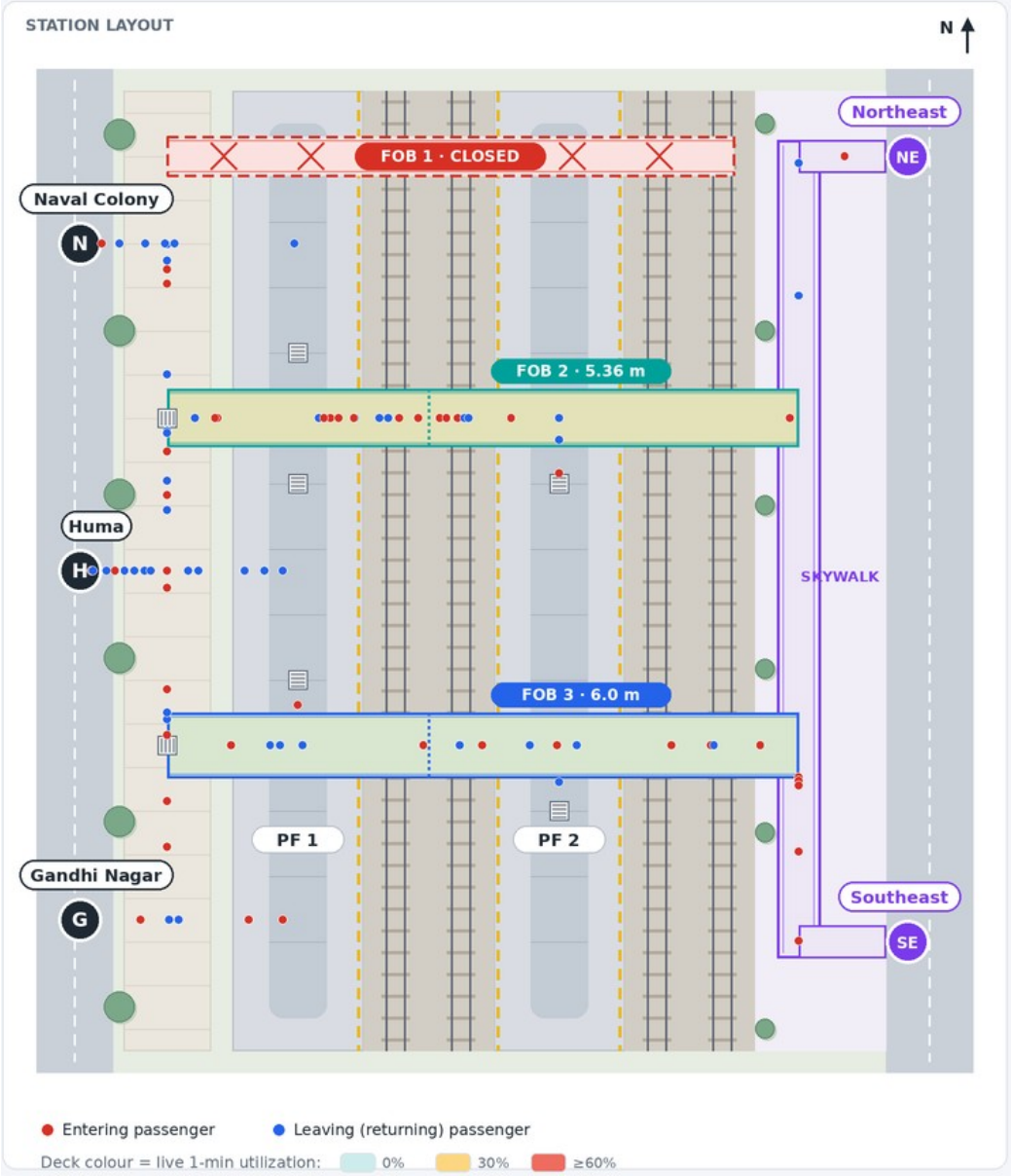


Figure 1. Simplified layout in the AnyLogic model (not to scale). Red = entering, blue = leaving.

Table 2. Model scope.

Included	Excluded
Five gates, entering and leaving streams + 2 extra-demand streams	Ticketing, buses, autos, metro operations
Routes: direct PF1/PF1A, FOB 2, FOB 3, skywalk → FOB 3	Stair/escalator capacity, density, queuing
FOB 2 and FOB 3 one-minute utilization	Skywalk utilization (not an FOB)

4. Observed Data and Input Parameters

Table 3. Baseline gate demand (pax/min).

Gate	Entering	Leaving	Total	Status
Naval Colony	10.00	42.00	52.00	Observed
Huma	27.20	50.00	77.20	Observed
Gandhi Nagar	33.33	6.66	39.99	Observed
Northeast	10.00	5.00	15.00	Assumed: 50% of East-side count
Southeast	10.00	5.00	15.00	Assumed: 50% of East-side count
Total	90.53	108.66	199.19	

Source: IE 301 project report, Section 2.2.

Table 4. Observed FOB flows and widths (benchmark only).

Facility	Observed flow (pax/min)	Width (m)	Width status
FOB 1	Closed	–	Not in use
FOB 2	70.0	5.36	Recorded; to be verified
FOB 3	33.3	6.0	Unverified

Source: project report Sections 2.3 and 4.6.

5. Gate Route-Choice Assumptions

Route splits are **modelling assumptions, not measured choices**. The same split applies to entering and leaving passengers at a gate.

Table 5. Route-choice probabilities by gate.

Gate	Direct PF1/PF1A	Via FOB 2	Via FOB 3	Skywalk → FOB 3
Naval Colony	60%	25%	15%	0%
Huma	55%	30%	15%	0%
Gandhi Nagar	60%	35%	5%	0%
Northeast	0%	95%	0%	5%
Southeast	0%	30%	70%	0%
Huma (additional demand)	50%	35%	15%	0%
Gandhi Nagar (additional demand)	40%	40%	20%	0%

Last two rows apply to the additional demand only. Skywalk users are counted on FOB 3.

5.1 Route-split noise

Each replication draws each gate's split from a Dirichlet distribution around these values (κ = 99, about ±3 pp for a 10–15% share). Zero shares stay zero.

$$(p_PF1,\ p_FOB2,\ p_FOB3,\ p_sky) \sim Dirichlet(\kappa \cdot \bar{p}), \quad \kappa = 99 \tag{5.1}$$

5.2 Expected flows vs observed counts

Table 6. Expected baseline flows (no randomness) vs observed.

Facility	Expected (pax/min)	Expected U	Observed (pax/min)	Observed U
FOB 2	68.9	21.4%	70.0	21.8%
FOB 3	32.6	9.1%	33.3	9.3%

FOB 2 = 13.0 + 23.2 + 14.0 + 14.25 + 4.5; FOB 3 = 7.8 + 11.6 + 2.0 + 10.5 + 0.75 (skywalk).

> **Not validation.** The splits were chosen as estimates; their agreement with observed counts is a consistency check only.

6. Ghatkopar-Referenced Demand Scenarios

$$+50\ pax/min = 500 \times 0.10, \qquad +100\ pax/min = 500 \times 0.20 \tag{6.1}$$

Table 7. Additional entering demand by gate (pax/min).

Scenario	Huma (80%)	Gandhi Nagar (20%)	Total
Baseline	0	0	0
+50 pax/min	+40	+10	+50
+100 pax/min	+80	+20	+100

500 pax/min Ghatkopar flow is assumed. Baseline rates are unchanged.

Table 8. Expected routing of the additional demand (pax/min).

Route	+50 pax/min	+100 pax/min
Direct PF1/PF1A	+24	+48
FOB 2	+18	+36
FOB 3	+8	+16

Huma extra × (50/35/15); Gandhi Nagar extra × (40/40/20).

7. Stochastic Arrival and Route-Choice Model

Table 9. Distributions.

Element	Distribution	Parameters
Arrivals (12 streams)	Poisson process	λ = gate rate; extra streams 0.8·Δ and 0.2·Δ
Inter-arrival time	Exponential	Mean 60/λ s
Gate split per replication	Dirichlet	κ = 99
Route per passenger	Categorical (inverse CDF)	That replication's split
Walking speed	Deterministic	1.30 m/s

$$\Delta t = -\ln(1 - U) \cdot 60/\lambda, \quad U \sim Uniform(0, 1) \tag{7.1}$$

A crossing is counted once, in the minute the passenger reaches the deck mid-point. 95% CI over n = 30: $\bar{x} \pm 2.045 \cdot s/\sqrt{30}$. Seeds 1001–1030 are reused in every scenario (common random numbers). Empty start, no warm-up.

8. Capacity and Utilization Calculations

$$C = W \times 60\ pax/m/min, \qquad U = Q \,/\, C \times 100\% \quad (8.1)$$

Table 10. Reference capacities.

Facility	W (m)	C (pax/min)
FOB 2	5.36	321.6
FOB 3	6.0 (unverified)	360.0

Limit. U is a reference-capacity ratio, not a level-of-service or safety measure; stairs, counter-flow and platoons are not modelled.

9. AnyLogic Implementation

Table 11. Model structure (Kanjurmarg_FOB_Simulation.alp, AnyLogic 8.8).

Element	Role
Arrival(stream)	Poisson arrival, route choice, new passenger
Waypoint / Crossing	Move passengers; count FOB crossings per minute
minuteTick (60 s)	Utilization, gauges, deck colours, time plot
runBatch()	30 replications × 3 scenarios; results to console and CSV
gateSplits, extraSplits, splitNoiseKappa	Editable route splits and noise

Screen: station layout (decks coloured by live utilization), FOB 2 and FOB 3 utilization gauges (now, mean, peak) and a utilization-over-time plot.

Verification. Model code compiles against AnyLogic API stand-ins; its headless run reproduced replication 1 exactly (4,157 FOB 2 and 2,058 FOB 3 crossings). Opening in AnyLogic itself is still to be confirmed.

10. Experimental Design

Table 12. Experiments.

Factor	Setting
Scenarios	Baseline, +50, +100 pax/min
Runs	60 min × 30 replications per scenario, seeds 1001–1030
Route-split noise	On (κ = 99); a fixed-split set (κ = 0) is used only to size its effect

11. Simulation Results

Table 13. One-minute utilization (%), 30 replications.

Scenario	FOB	Mean ± 95% CI	P95 minute	Busiest minute
Baseline	FOB 2	20.7 ± 0.5	24.8	32.3
Baseline	FOB 3	9.3 ± 0.4	12.0	16.9
+50 pax/min	FOB 2	26.3 ± 0.5	31.1	37.6
+50 pax/min	FOB 3	11.6 ± 0.4	14.5	19.7
+100 pax/min	FOB 2	31.9 ± 0.6	37.2	45.1
+100 pax/min	FOB 3	13.8 ± 0.5	17.1	23.1

Source: data/scenario_summary.csv.

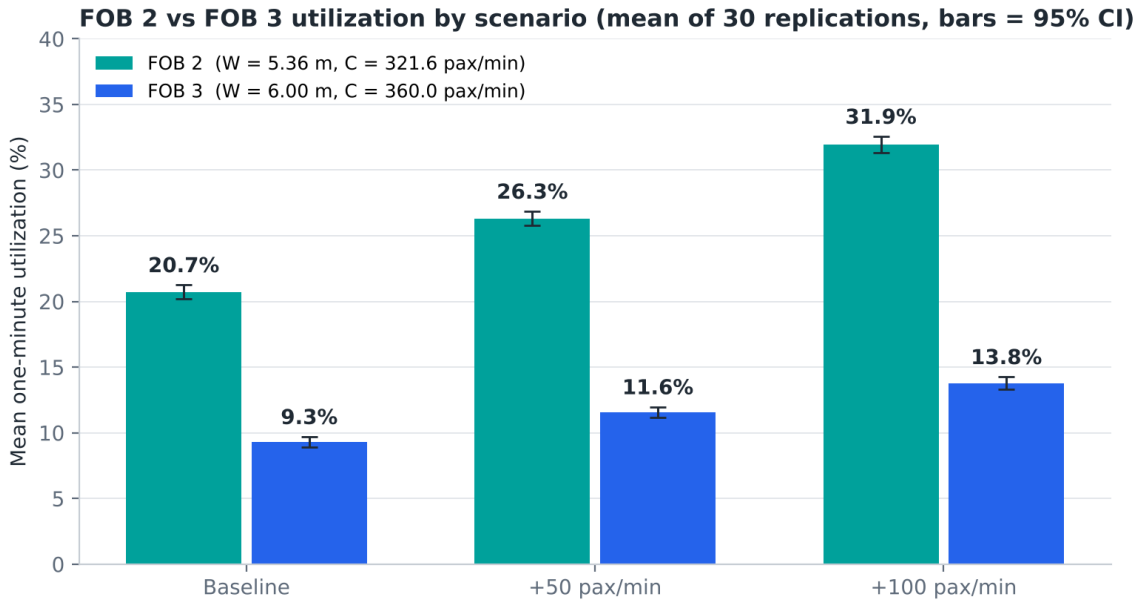


Figure 2. FOB 2 vs FOB 3 mean utilization by scenario.

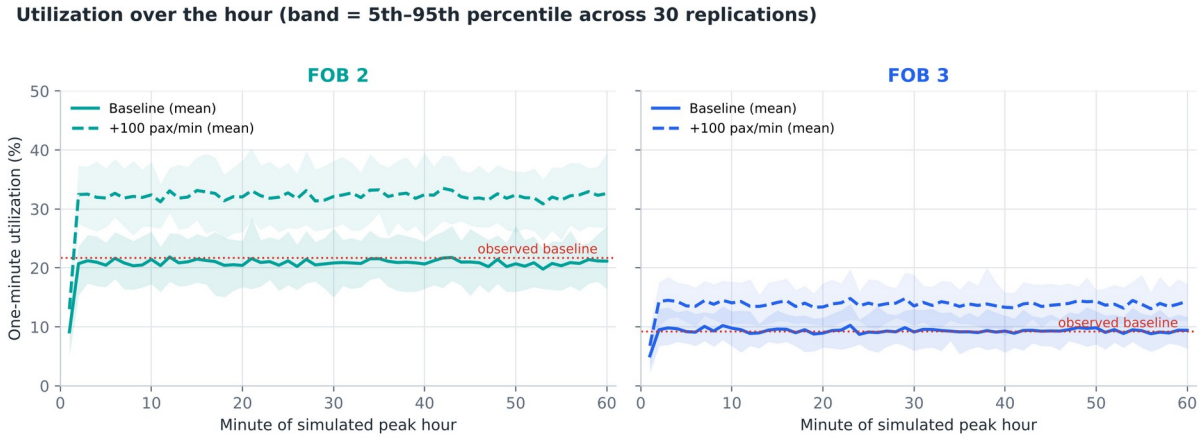


Figure 3. Utilization over the hour, baseline vs +100 pax/min (band = 5th–95th percentile).

About 94–96% of the replication-to-replication variance comes from route-split noise rather than random arrivals.

12. Interpretation

1. **Demand split:** 49% direct PF1/PF1A, 35% FOB 2, 16% FOB 3, 0.4% skywalk.
2. **Baseline:** FOB 2 20.7%, FOB 3 9.3%.
3. **Extra demand:** +5.6 / +11.2 points on FOB 2 and +2.3 / +4.5 on FOB 3 for +50 / +100 pax/min.
4. **Variability:** busiest minutes are about 30% (FOB 2) and 43% (FOB 3) above the hourly mean.
5. **Observed counts:** simulated baseline (20.7%, 9.3%) is close to observed (21.8%, 9.3%) — a consistency check, not validation.
6. **Balance:** FOB 2 carries 67% of FOB flow at baseline and 67% at +100 pax/min.

> **Implication.** Deck width is not binding even at +100 pax/min; FOB 2 is the bridge to watch. Measured route-choice data would sharpen every result.

13. Assumptions, Limitations, and Additional Data Required

- East-side counts split 50/50 between Northeast and Southeast.
- Gate and additional-demand route splits are assumptions, not measurements.
- 500 pax/min Ghatkopar flow is assumed; +50/+100 are scenario inputs, not forecasts.
- FOB 3 width (6.0 m) is unverified; FOB 2 width (5.36 m) to be confirmed.
- Poisson arrivals ignore train-release platoons, so peak minutes are likely understated.

Table 14. Assumptions register.

ID	Assumption	Value	Evidence status
A1	Five access gates	Naval Colony, Huma, Gandhi Nagar (west); Northeast, Southeast (east, skywalk exits)	Layout decision
A2	East-side gate rates	NE and SE each 10.00 entering / 5.00 leaving pax/min	ASSUMPTION (50/50 split)
A3	Baseline gate route-choice probabilities	NC 60/25/15/0, Huma 55/30/15/0, GN 60/35/5/0, NE 0/95/0/5, SE 0/30/70/0 (PF1/FOB2/FOB3/skywalk)	Modelling assumption, not measured
A3b	Route shares of the additional demand	Huma 50/35/15, Gandhi Nagar 40/40/20 (PF1/FOB2/FOB3)	Modelling assumption, not measured
A4	Southeast split	70% FOB 3, 30% FOB 2	Explicit ASSUMPTION
A5	Leaving passengers	Use the same route percentages as entering passengers at that gate	Assumption
A6	Route-split noise	Dirichlet draw per gate per replication, concentration 99 (about +/-3 pp for a 10% share)	Modelling choice
A7	Skywalk	Connector between FOB 2 and FOB 3 on the east side; 5% of Northeast users walk it to FOB 3	Layout decision
A8	FOB 1	Closed; never offered as a route	Observed

ID	Assumption	Value	Evidence status
A9	Ghatkopar reference flow	500 pax/min	ASSUMED, not independently verified
A10	Additional demand	+50 and +100 pax/min entering only; 80% Huma, 20% Gandhi Nagar; own route shares (A3b)	Scenario input, not a forecast
A11	Arrival process	Independent Poisson processes (exponential inter-arrival times) for 10 baseline + 2 additional streams	Simplification
A12	Constant rates over 60 min	Peak-period observed rates	Simplification
A13	FOB 2 width	5.36 m	Recorded; clear width unverified
A14	FOB 3 width	6.0 m	UNVERIFIED assumption
A15	Flow intensity I	60 pax/m/min	Reference value
A16	Walking speed / crossing point	1.30 m/s; counted at FOB deck mid-point	Assumption
A17	Initial state	Station starts empty, no warm-up	Assumption
A18	Replications and seeds	30 per scenario, seeds 1001-1030, common random numbers	Design choice

Full register: data/assumptions_register.csv.

Data needed: gate-to-route counts; separate Northeast/Southeast counts; synchronised gate and FOB counts; verified clear widths; one-minute profiles with train times; a verified Ghatkopar source.

14. Conclusion

Under the stated assumptions, FOB 2 runs at about 21% and FOB 3 at about 9% of reference capacity at baseline, rising to about 32% and 14% with +100 pax/min. FOB 2 remains the more loaded bridge. Measured route-choice data are the key next step.

15. Reproducibility Appendix

15.1 Deliverables

Table 15. Project files (folder kanjurmarg_fob_simulation/).

Path	Content
anylogic/Kanjurmarg_FOB_Simulation/ Kanjurmarg_FOB_Simulation.alp	Native AnyLogic 8.8 project
engine/FobEngine.java	Stochastic core (embedded verbatim in the .alp)
engine/RunnerMain.java, scripts/run_engine.sh	Stand-alone batch runner
python/fob_twin.py	Independent Python re-implementation (cross-check)
python/analyze.py, python/make_charts.py	CSV export, verification checks, utilization charts
scripts/build_alp.py, check_alp_code.py, render_preview.py	Model generator, model-code check, screen preview
data/*.csv	observed_data, gate_route_splits, expected_route_allocation, scenario_summary,

Path	Content
	fob_time_series, replication_results, route_split_uncertainty, assumptions_register
charts/*.png	All figures in this report
README.md, VALIDATION.md	Instructions and verification record

15.2 Reproducing the results

```
scripts/run_engine.sh 30 60 1.30 1001 99      # replications, minutes, walk speed, base seed, noise kappa
python3 python/analyze.py                      # CSVs + verification checks
python3 python/make_charts.py                  # utilization figures
python3 scripts/build_alp.py && python3 scripts/check_alp_code.py
```

In AnyLogic: open the .alp from a folder AnyLogic can read (on macOS not Downloads, unless AnyLogic is allowed access), run the Simulation experiment, choose a scenario and press Run. 'Run replications' runs 30 seeds × 3 scenarios inside AnyLogic and reproduces data/scenario_summary.csv; output goes to anylogic_batch_results.csv in the model folder.

15.3 Verification summary

Table 16. Automated checks.

Check	Result
Route probabilities sum to 100% for every gate (defaults)	PASS — Naval Colony 100%; Huma 100%; Gandhi Nagar 100%; Northeast 100%; Southeast 100%
Additional-demand route shares sum to 100%	PASS — Huma 50/35/15, Gandhi Nagar 40/40/20
Route probabilities sum to 100% for every gate (noisy draws)	PASS — checked 90 replications x (5 gates + 2 additional-demand splits)
Impossible routes stay impossible under noise	PASS — routes with 0% default probability are never used
FOB 1 remains unavailable	PASS — FOB 1 crossings over 180 replications = 0
Skywalk is a connector, not an FOB	PASS — only Northeast passengers use the skywalk (8361 in total); they are counted on FOB 3 and there is no skywalk crossing count
Direct PF1 passengers never counted; each FOB passenger counted at most once	PASS — every replication: FOB 2 + FOB 3 crossings + passengers still approaching at t=60 = FOB-route passengers
East-side rates conserve the observed aggregate	PASS — Northeast + Southeast entering = 10 + 10 = 20; leaving = 5 + 5 = 10 pax/min
Baseline gate totals	PASS — entering 90.53, leaving 108.66, total 199.19 pax/min
Additional demand total and 80/20 split (+50 pax/min)	PASS — Huma +40, Gandhi Nagar +10, total +50 pax/min; Huma entering 67.20, Gandhi Nagar entering 43.33
Additional Huma arrivals are Poisson (+50 pax/min)	PASS — mean 2406.9 per hour vs expected 2400 (z = +0.77)
Additional Gandhi Nagar arrivals are Poisson (+50 pax/min)	PASS — mean 598.3 per hour vs expected 600 (z = -0.39)
Additional demand total and 80/20 split (+100 pax/min)	PASS — Huma +80, Gandhi Nagar +20, total +100 pax/min; Huma entering 107.20, Gandhi Nagar entering 53.33
Additional Huma arrivals are Poisson	PASS — mean 4792.5 per hour vs expected 4800 (z = -0.60)

Check	Result
(+100 pax/min)	
Additional Gandhi Nagar arrivals are Poisson (+100 pax/min)	PASS — mean 1193.4 per hour vs expected 1200 (z = -1.04)
Expected additional routing +50	PASS — PF1 +24, FOB 2 +18, FOB 3 +8 pax/min (expected values, reported separately from simulation)
Expected additional routing +100	PASS — PF1 +48, FOB 2 +36, FOB 3 +16 pax/min (expected values, reported separately from simulation)
Baseline streams unchanged by scenarios (no multipliers)	PASS — with the same seed, all 10 baseline gate-direction streams generate identical arrivals in every scenario
Poisson arrival counts for the 10 baseline streams	PASS — z-scores of hourly means vs lambda x 60: +0.2, +0.9, +0.9, +1.4, +0.5, -0.6, -1.3, -0.9, +0.8, -2.5
Per-passenger route choice follows each replication's split	PASS — pooled chi-square = 946.5 on 960 df, p = 0.616 (1,077,324 passengers)
Route-split noise has the intended size	PASS — SD of Naval Colony FOB 3 share (15%) across replications = 3.5 pp (theory 3.6 pp)
Different seeds generate different trajectories	PASS — 30 distinct arrival-count vectors from 30 seeds
Java engine reproduced by independent Python implementation	PASS — identical per-minute FOB 2 counts for sampled replications (full comparison of all 180 replications: 0 mismatches)
Capacity units	PASS — C2 = 5.36 m x 60 pax/m/min x 1 min = 321.6 pax; C3 = 6.0 x 60 x 1 = 360 pax
Simulated results recorded separately from expected values	PASS — baseline simulated minus expected flow: FOB 2 -2.25 pax/min, FOB 3 +0.81 pax/min
Java engine vs Python twin, all 180 replications	PASS — 0 mismatches in arrivals, route counts, noisy splits and per-minute crossings
Model code compile (scripts/check_alp_code.py)	PASS — all functions, events and 14 dynamic expressions compile against AnyLogic API stand-ins
Model event logic executed headless	PASS — identical to batch replication 1 (seed 1001); in-model replications reproduce scenario_summary.csv
Opened and run in AnyLogic 8.8	NOT YET CONFIRMED — to be checked on the user's computer

15.4 Software

Java (OpenJDK 21), Python 3.13 with NumPy 2.5, pandas 3.0, SciPy 1.18 and Matplotlib 3.11. Target: AnyLogic 8.8 (Personal Learning Edition compatible; no paid libraries).